TREKKING BUSINESS ANALYTICS

Project Overview

This project focuses on analyzing trekking business data to understand booking trends, revenue patterns, cancellation behavior, customer demographics, and seasonal performance.The objective is to generate actionable business insights that support:Revenue growth,Pricing strategy optimization,Cancellation risk management,Customer segmentation,Seasonal demand planning.

Dataset Summary

- Rows: 500
- Columns: 18
- Time Period: 2023–2025

Key Features:

1) **Customer Information:** Customer Email, Customer Age, Age Group, Customer City
2) **Booking Details:** Booking ID, Trek Location, Package Duration (Days), Difficulty Level, Season, Booking Date
3) **Financial Information:** Price Per Person, Group Size, Total Amount, Cancellation Status, Number of People Cancelled, Actual Revenue (After Refund Adjustments)

Data Preparation Using Python (Pandas)

1. Data Cleaning

- Converted booking_date to datetime

```
[6]: #change datatypes
     df["booking_date"] = pd.to_datetime(df["booking_date"])
```

- Checked for null values

```
[4]: df.isnull().sum()

[4]: booking_id                    0
     customer_email                0
     trek_location                 0
     package_duration_days         0
     difficulty_level              0
     booking_date                  0
     season                        0
     price_per_person              0
     group_size                    0
     total_amount                  0
     cancellation_status           0
     number_of_people_cancelled    0
     customer_age                  0
     customer_city                 0
     dtype: int64
```

2. Feature Engineering

- Created age_group column

```
[9]: # age group column
bins = [17, 24, 34, 100]
labels = ["Young Adult", "Adult", "Senior"]

df["age_group"] = pd.cut(
    df["customer_age"],
    bins=bins,
    labels=labels
)
```

- Created actual_revenue column using cancellation multiplier

```
#actual_revenue
refund_map = {
    "Completed": 1,
    "Partial Cancellation": 0.5,
    "Full Refund": 0
}

df["actual_revenue"] = (
    df["total_amount"] *
    df["cancellation_status"].map(refund_map)
).fillna(0)
```

- Calculated Cancellation Rate

```
[11]: #cancellation rate
cancellation_rate = (
    df["cancellation_status"].str.contains("Cancelled").sum()
    / len(df)
) * 100
print(cancellation_rate)

14.6
```

- Extracted month_name and year

```
[8]: #create columns for year and month
df["year"] = df["booking_date"].dt.year
df["month"] = df["booking_date"].dt.month_name()
```

3. Data Analysis

- Actual revenue by location

```
[15]: #Group-wise Analysis
df.groupby("trek_location")["actual_revenue"].sum().sort_values(ascending=False)

[15]: trek_location
Kedarkantha          4993100.0
Sandakphu            4447500.0
Hampta Pass          3093000.0
Valley of Flowers    2287350.0
Hampi                2260350.0
Triund                987800.0
Name: actual_revenue, dtype: float64
```

- Most preferred trek

```
[16]: #Most prefered trek
      df["trek_location"].value_counts()
```

```
[16]: trek_location
      Kedarkantha          106
      Sandakphu             92
      Hampi                 87
      Triund                84
      Hampta Pass           69
      Valley of Flowers     62
      Name: count, dtype: int64
```

- Cancellation By Season

```
[17]: #Cancellation by Season
      df.groupby("season")["number_of_people_cancelled"].sum()
```

```
[17]: season
      Moderate    94
      Off         69
      Peak        69
      Name: number_of_people_cancelled, dtype: int64
```

4. <u>Database Integration</u>

- Connected Python to PostgreSQL using SQLAlchemy

```
#Connect python to pgadmin for further analysis
import pandas as pd
from sqlalchemy import create_engine

# Step 1: Connect to PostgreSQL
username="postgres"
password="Admin"
host="localhost"
port="5433"
database="Trek_Business_Analysis"

engine=create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}")

# Step 2: Load DataFrame into PostgreSQL
table_name="trek"
df.to_sql(table_name, engine, if_exists="replace", index=False)

print(f"Data successfully loaded into table '{table_name}' in database '{database}'.")
```

```
Data successfully loaded into table 'trek' in database 'Trek_Business_Analysis'.
```

- Loaded cleaned data into PostgreSQL

```
-- check the data
select * from trek;
```

	booking_id bigint	customer_email text	trek_location text	package_duration_days bigint	difficulty_level text	booking_date timestamp without time zone	season text	price_per_person bigint	group_size bigint	total_amo bigint
1	1	user1@gmail.com	Hampta Pass	5	Moderate	2023-10-26 00:00:00	Peak	14400	4	
2	2	user2@gmail.com	Sandakphu	6	Difficult	2025-03-15 00:00:00	Moderate	15000	3	
3	3	user3@gmail.com	Kedarkantha	7	Moderate	2023-09-06 00:00:00	Moderate	14000	6	
4	4	user4@gmail.com	Triund	2	Easy	2024-02-23 00:00:00	Peak	4800	2	
5	5	user5@gmail.com	Hampi	5	Easy	2023-10-16 00:00:00	Peak	10800	3	
6	6	user6@gmail.com	Hampta Pass	5	Moderate	2025-04-05 00:00:00	Moderate	12000	5	
7	7	user7@gmail.com	Hampi	5	Easy	2023-09-12 00:00:00	Moderate	9000	1	
8	8	user8@gmail.com	Hampi	5	Easy	2023-11-29 00:00:00	Peak	10800	2	

Performed structured SQL analysis

1) Top 3 Most Profitable Treks

```
-- Top 3 Most Profitable Treks
select trek_location,sum(actual_revenue) as total_revenue
from trek
group by trek_location
order by total_revenue desc
limit 3;
```

	trek_location text	total_revenue double precision
1	Kedarkantha	4993100
2	Sandakphu	4447500
3	Hampta Pass	3093000

2) Highest Cancellation Season

```
--Highest Cancellation Season
select season,sum(number_of_people_cancelled) as cancelled_people
from trek
group by season
order by cancelled_people desc
limit 1;
```

	season text	cancelled_people numeric
1	Moderate	94

3) Year-wise revenue growth

```
--Year wise revenue growth
select year,sum(actual_revenue) as total_revenue
from trek
group by year
order by year;
```

	year integer	total_revenue double precision
1	2023	5371350
2	2024	6593700
3	2025	6104050

4) Average booking value per city

```
--Average booking value per city
select customer_city,
round(avg(total_amount)::numeric, 2) as avg_revenue
from trek
group by customer_city
order by avg_revenue desc;
```

	customer_city text	avg_revenue numeric
1	Chennai	43457.89
2	Delhi	43294.12
3	Hyderabad	42558.25
4	Mumbai	42181.51
5	Bangalore	39532.05
6	Pune	34829.17

5) Peak Season Price impact

```
-- Peak season price impact
select season,
avg(price_per_person) as avg_price,
sum(actual_revenue) as total_revenue
from trek
group by season
order by avg_price desc;
```

	season text	avg_price numeric	total_revenue double precision
1	Peak	13491.089108910891	8352000
2	Moderate	11580.110497237569	6278000
3	Off	9081.1965811965811966	3439100

6) Revenue loss due to cancellations

```
--Revenue loss due to cancellations
select sum(total_amount - actual_revenue) as total_revenue_loss from trek;
```

	total_revenue_loss double precision
1	2538000

7) Total cancellation Rate

```
--Cancellation rate
select
round(sum(number_of_people_cancelled)::numeric /sum (group_size) * 100,2) as cancellation_rate
from trek;
```

	cancellation_rate numeric
1	13.28

8) Total revenue by age group

```
--Revenue by age group
select age_group,sum(actual_revenue) as total_revenue
from trek
group by age_group
order by total_revenue desc;
```

	age_group text	total_revenue double precision
1	Senior	10570650
2	Adult	4086500
3	Young Adult	3411950

9) Average group size by trek location

```
--Average Group size by Trek Location
select trek_location,round(avg(group_size)::numeric) as avg_group_size
from trek
group by trek_location
order by avg_group_size;
```

	trek_location text	avg_group_size numeric
1	Sandakphu	3
2	Hampi	3
3	Triund	3
4	Hampta Pass	4
5	Valley of Flowers	4
6	Kedarkantha	4

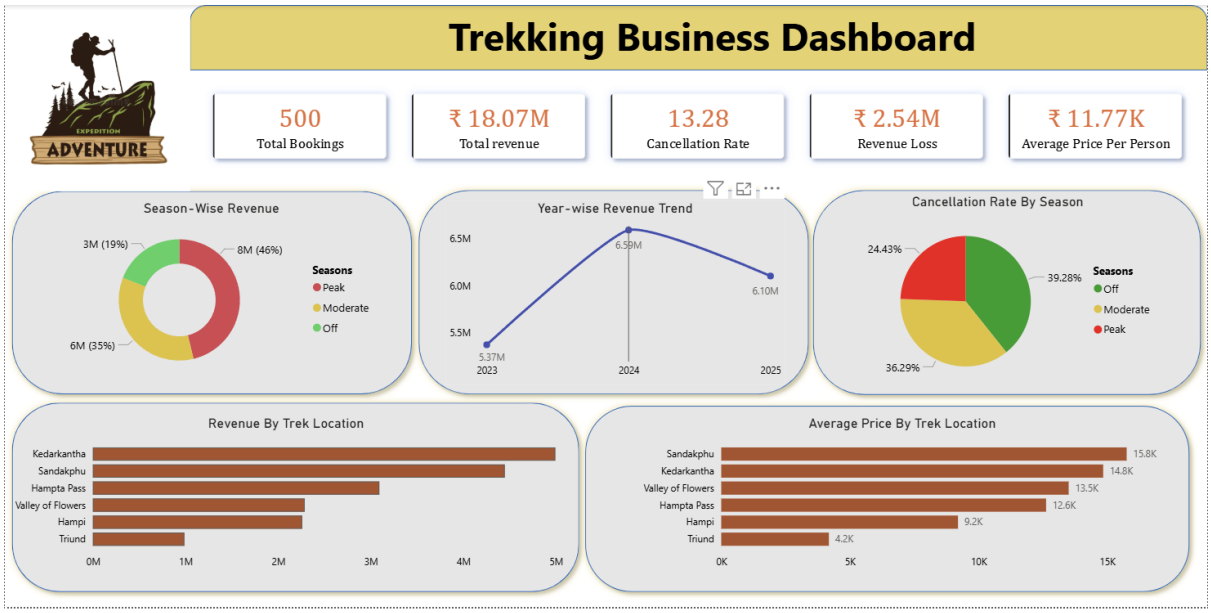
10) Total revenue by season

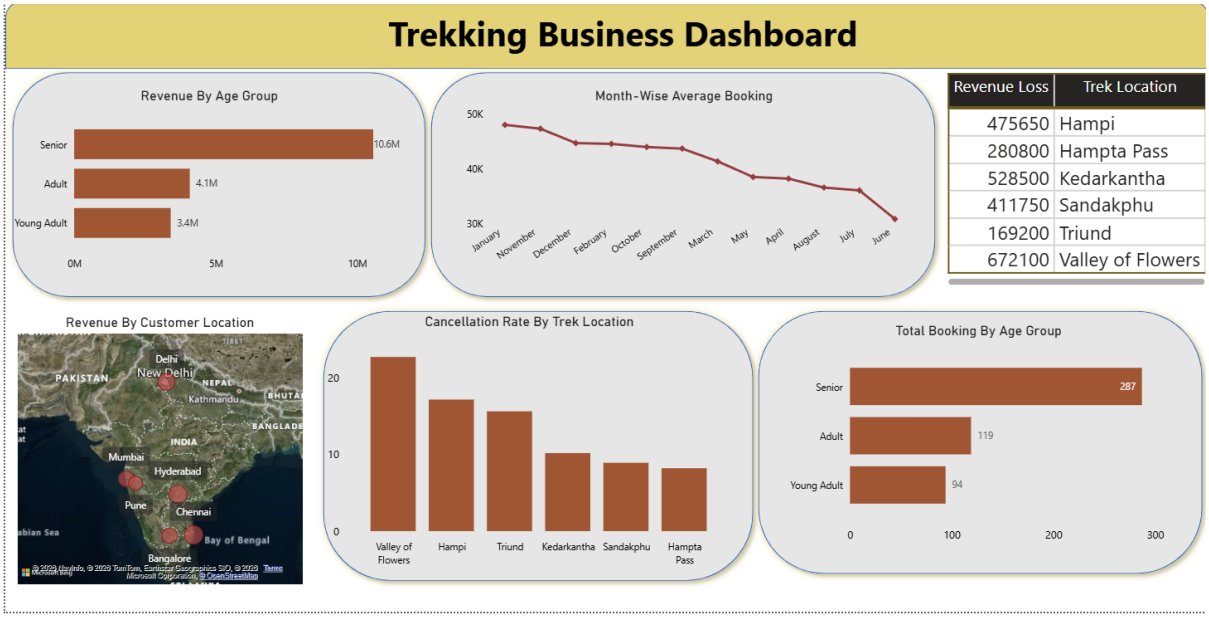
```
--Revenue by season
select season,
round(sum(actual_revenue)::numeric,2) as total_revenue
from trek
group by season
order by total_revenue desc;
```

	season text	total_revenue numeric
1	Peak	8352000.00
2	Moderate	6278000.00
3	Off	3439100.00

Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.





📌 Key Business Insights

✔ Peak season drives maximum revenue
✔ Longer packages generate higher total revenue but higher cancellation risk
✔ Certain cities are high-value markets
✔ Revenue leakage due to cancellations impacts profitability
✔ Medium-duration packages offer best revenue efficiency

📌 Business Recommendations

✓ Strengthen cancellation policy during off-season
✓ Introduce flexible pricing strategies
✓ Increase marketing in top revenue cities
✓ Promote medium-duration treks for better efficiency
✓ Offer loyalty incentives for repeat customers